

Question number	Answer	Mark
3(a)(i)	An answer showing the following steps: • change in mass calculated (1) • percentage change calculated (1) Example of calculation: $2.75 - 1.92 = -0.83 \text{ g}$ $(-0.83 \div 2.75) \times 100 = -30.18\%$	(2)

Question number	Answer	Mark
3(a)(ii)	A graph showing the following features: • A axes correct (x – concentration of sucrose solution, y - percentage change in mass) (1) • L axes correctly labelled, and with units mol dm^{-3} and % (1) • P correct plotting (1) • S points joined with straight lines (1) Example of graph:	(4)

Question number	Answer	Additional guidance	Mark
3(a)(iii)	An answer showing the following steps: • 0.35 (1) • mol dm^{-3} (1)	Allow ecf from graph	(2)

Question number	Answer	Mark
3(a)(iv)	An explanation that includes the following points: • there is no (net) change in the quantity of water in the carrot tissue / the water potential in the carrot equals the water potential of the solution (1) • {because / therefore} the rate of water gain is equal to its loss (1)	(2)

Question number	Answer	Mark
3(b)(i)	Any two from the following: • surface area / volume / age / variety / storage conditions / source (of beetroot) / same {wavelength / filter} (1)	(2)

Question number	Answer	Mark
3(b)(ii)	• intensity of the red colour	(1)

Question number	Answer	Mark
3(b)(iii)	A graph showing the following features: - axes correctly labelled with units (x temperature/ °C and y intensity of the red colour/a.u.) (1) - scales correctly labelled (1) - standard deviations (SDs) all correctly plotted (1) Example graph:	(3)

Question number	Answer	Additional guidance	Mark
3(b)(iv)	An answer that includes the following points: {cells / membranes / eq} damaged (by cutting up of pieces) (1) - so pigment could leak out of {vacuoles / cells} (1)	Accept reference to condensation on cuvette at low temperature, leads to absorption of some of the light	(2)

Question number	Answer	Mark
3(b)(v)	An answer that includes any five of the following points: • (overall) the intensity of the red colour increases as temperature increases (1) • but from 0 °C to 40 °C, the SDs overlap so no significant effect (1) • at 50 °C the mean is higher than that at 40 °C but SDs overlap so there is no case for saying the difference is significant (1) • at 60 °C the mean is higher than that at 50 °C and the SDs do not overlap so this difference can be regarded as significant (1) • at 70 °C the mean degree of redness falls from that at 60 °C but the SDs overlap so temperatures above 60 °C appear to have no further effect (1) • there are no data above 70 °C so cannot say what any further rise in temperature would cause (1)	(5)